

I'm as Mad as Hell, and I'm Not Going to Take Physical Types Anymore!

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What is Vortex?

- “An extensible, state-of-the-art framework for columnar compression”
- Linux Foundation project
- Started @ SpiralDB, now adopted by many others.

- Columnar file format (a la Parquet)
- Compute on compressed data
- Highly extensible and flexible
- Integrates cutting edge research - FSST, OnPair, ALP, FastLanes
- Has a Python (including Arrow Datasets), DuckDB extension, Iceberg (pending), and of course - DataFusion.

There is no Utf8.
There is no LargeUtf8.
There is no Utf8View.

There is only String.

- An array's type doesn't (fully) determine how its physically stored.
- Primitive, Binary, String, List, etc.
- Arrays ~ Lazy compute graph

Arrow vs. Vortex types - not a 1:1 mapping

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	Arrow	Vortex
Strings	Utf8 / LargeUtf8 / Utf8View core type	Utf8 core type
Lists	List / LargeList / ListView / LargeListView core type	List core type
Decimals	Decimal32 / 64 / 128 / 256 core type width is part of the type	Decimal(precision, scale) core type width is a storage detail
Dictionary / RLE	Dictionary<K, V> / RunEndEncoded core type	dict / runend encoding stripped on import, picked on export
Date & time	Date32 / Date64 / Time32 / Time64 / Timestamp core type	vortex.date / vortex.time / vortex.timestamp extension over Primitive storage
Variant	arrow.parquet.variant extension over Struct<metadata, value?, typed_value?>	Variant core type parquet.variant is one of its encoding

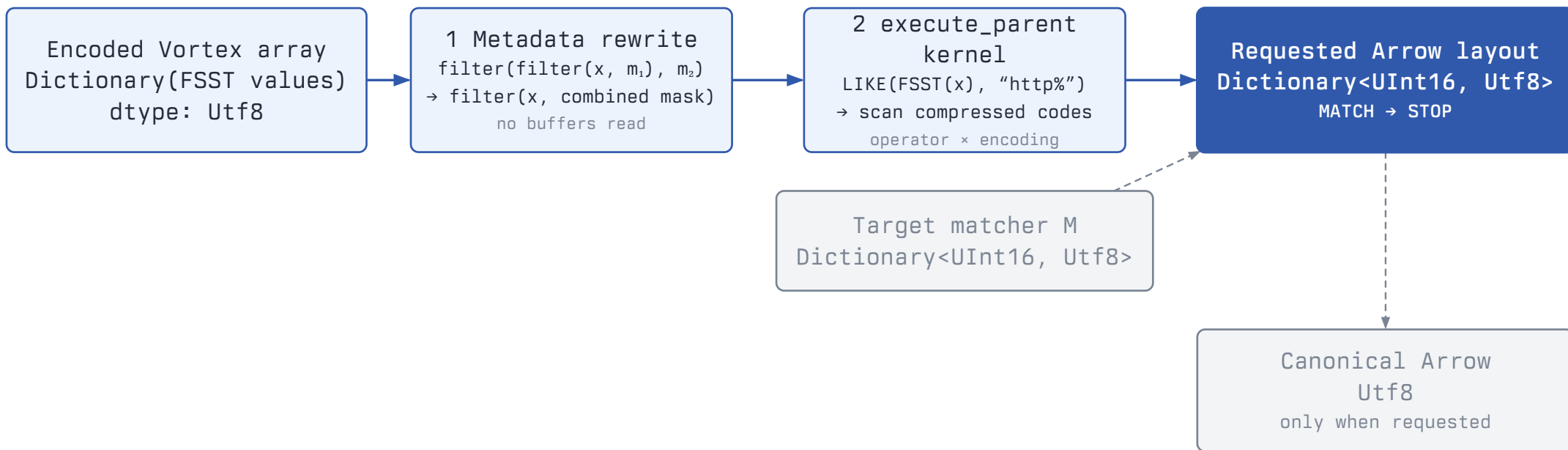
Decompress to the canonical layout, then hand Arrow the same buffer.



Docs: <https://docs.vortex.dev/developer-guide/internals/execution>
Code: [vortex-arrow/executor/primitive.rs](#) · [vortex-buffer/arrow.rs](#)

Example #2 - Stop at the requested layout

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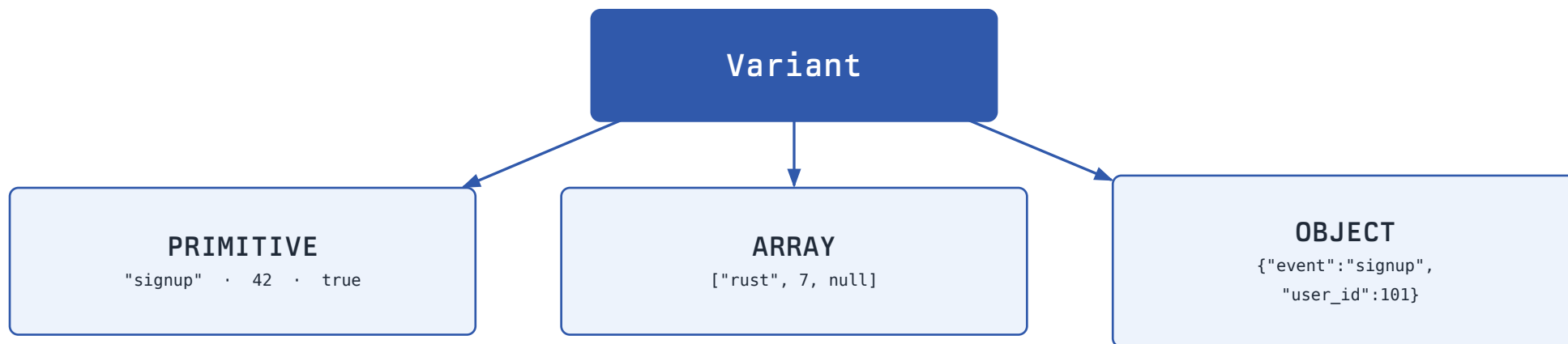


Execute only until the requested physical layout matches.

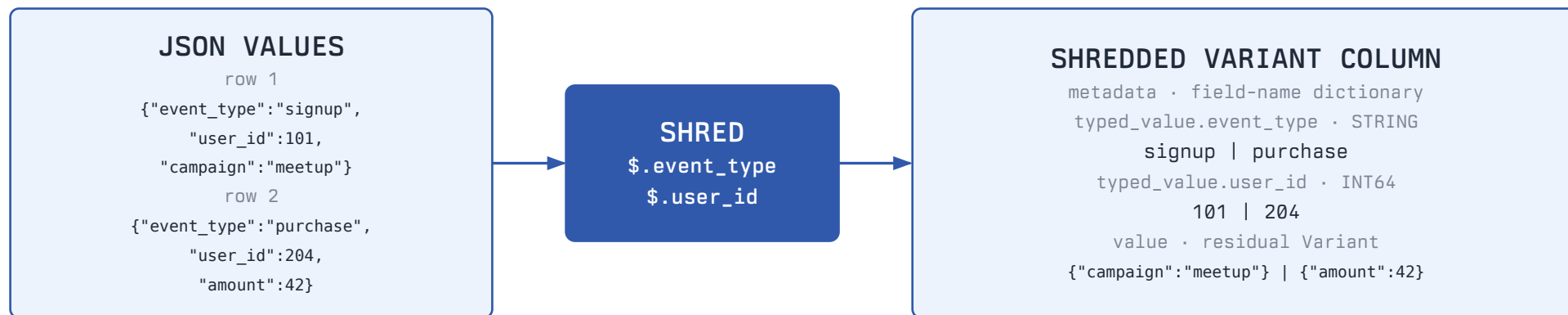
Docs: <https://docs.vortex.dev/developer-guide/internals/execution>
Code: `filter/rules.rs` · `fsst/compute/like.rs`

Variant

One logical type for semi-structured values.



The table schema stays Variant even when fields and shapes change.



Known fields become typed columns; everything else remains Variant.

VORTEX · ICEBERG · SPARK

shape is a storage detail

type: Variant

shredded paths live in the data, not the schema

ARROW · PARQUET

shape is part of the type

Struct<metadata, value?, typed_value?>

shred differently → different type

Keep Variant logical; shred known paths for columnar speed.

Thank You!

```
# Rust/DataFusion  
cargo add vortex # vortex-  
datafusion
```

```
# DuckDB  
INSTALL vortex;  
LOAD vortex;
```

```
# Python  
pip install vortex-data
```

<https://vortex.dev>

<https://vortex.dev/code>

<https://vortex.dev/slack>

<https://bench.vortex.dev>